TECHNICAL REPORT: NLP Analysis of "Arrival" (2016)

Assignment: A04 Analyzing "Arrival" Through the Lens of NLP

Course: ITAI 2376 - Deep Learning

Institution: Houston City College

Date: February 12, 2026

Team 1: DeMarcus Crump, Chloe Tu, Matthew Choo, Franck Eric Kolontchang Toheu

Executive Summary

This report analyzes *Arrival* (2016) through the lens of Natural Language Processing (NLP). The film's central conflict, trying to communicate with aliens who share zero common ground with humanity, turns out to be a surprisingly accurate dramatization of problems NLP researchers face every day: words with multiple meanings, languages with no training data, and communication systems that defy linear processing.

We map specific scenes to core NLP challenges including Word Sense Disambiguation, Zero-Shot Learning, and Multimodal Grounding. We evaluate the protagonist's methodology against rule-based, statistical, and deep learning paradigms, and reflect on the film's portrayal of the Sapir-Whorf hypothesis and its implications for AI alignment and dataset bias. A companion Google Colab notebook ("Project Heptapod") provides working code implementations of each concept analyzed in this report.

1. Introduction & Problem Statement

The story follows Dr. Louise Banks, a linguist recruited to decode "Heptapod B," the written language of alien visitors. These beings communicate through circular ink-like logograms that encode entire thoughts simultaneously, with no beginning or end. The core crisis comes when a Heptapod message gets translated as "weapon" by some nations and "tool" by others. One word, two interpretations, potential global war.

Importantly, the film establishes early on that translation is never straightforward, even between human languages. When Colonel Weber first recruits Louise, she explains that the Sanskrit word for "war" literally translates as "a desire for more cows." The original meaning has nothing to do with combat; it is an idiomatic expression whose cultural context has been lost in translation. This moment foreshadows the central crisis: if human languages already encode meaning in culturally specific, non-literal ways, the challenge of decoding an alien language with zero shared cultural reference points is exponentially harder.

Our team identified three objectives:

1. Map cinematic depictions of translation to specific NLP sub-domains.
2. Evaluate communication methods against rule-based, statistical, and deep NLP paradigms.
3. Reflect on the Sapir-Whorf hypothesis and its relevance to AI alignment and dataset bias.

2. Identified NLP Challenges – Research by Franck

2.1 Word Sense Disambiguation (WSD) and Ambiguity

The Crisis: The Heptapods transmit a message translated as "OFFER WEAPON." General Shang's team, which used competitive Mahjong as its teaching framework, interprets this as a threat. Dr. Banks hypothesizes it means "tool" or "gift."

NLP Parallel: This is a textbook case of **polysemy**, one word carrying multiple meanings. Think of "bank" (financial institution vs. riverbank). Humans resolve this from context instantly, but machines need explicit help. In deep learning, a word's embedding (a numeric representation of meaning) gets shifted by surrounding context to land on the right definition. Modern models like BERT do exactly what Louise does: argue that the surrounding context points to "tool," not "weapon."

To demonstrate this, we built a zero-shot classifier using BART that evaluates which interpretation best fits the message:

```
from transformers import pipeline
classifier = pipeline("zero-shot-classification", model="facebook/bart-large-mnli")
message = "The beings offer weapon to humanity for the future."
labels = ["military threat", "helpful gift or tool", "neutral transfer"]
result = classifier(message, labels)
# Result: 'helpful gift or tool' at 79.8% confidence
```

Without any Heptapod-specific training, the model picks up on contextual clues ("offer," "for the future") to favor "helpful gift or tool." The framework you use to teach language shapes the answers you get. China used Mahjong (competitive), so their model learned competitive concepts. Louise used collaborative conversation and got collaborative answers. In machine learning, this is training data bias, one of the biggest concerns in AI safety today.

The film also illustrates how sarcasm and tone create friction even between humans speaking the same language. Ian Donnelly, the physicist partnered with Louise, frequently uses dry humor and sarcastic remarks that the military personnel take at face value or dismiss entirely. When Ian quips about the Heptapods' first words being "a palindrome," it is a joke meant to highlight the strangeness of the situation, but in the tense military environment, that kind of figurative speech gets lost. This is a real NLP problem: sarcasm detection remains one of the hardest tasks in sentiment analysis because the literal meaning of the words is the opposite of the intended meaning. A model reading "great, another meeting" has no way to know if the speaker is genuinely enthusiastic or deeply annoyed without understanding tone, context, and speaker intent.

(Note: Figures 1-6 referenced in original document are omitted for text format) - By Matthew

Fig 1: Interpreting the Language

Fig 2: Analyzing the language

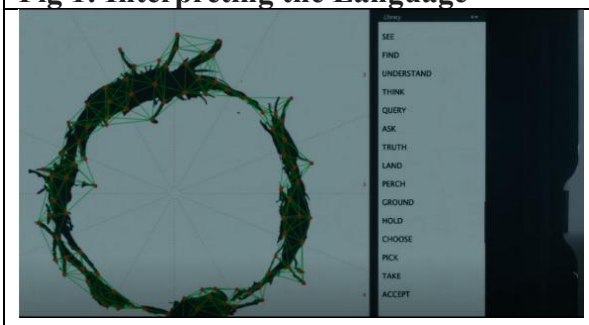

Fig 3: Translating

Fig 4: Thousand Words

Fig 5: Time 1/12

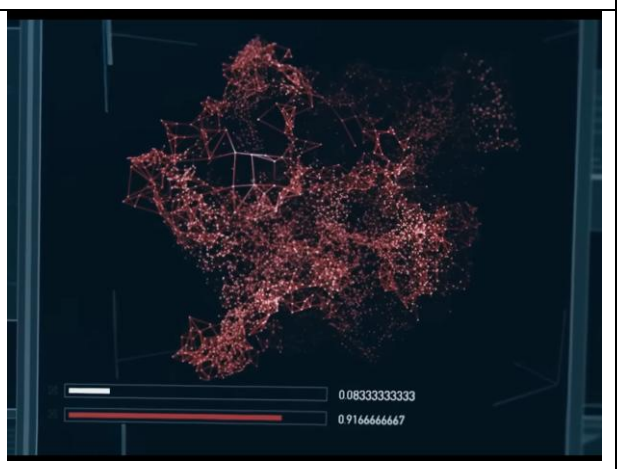

Fig 6: Trying to Interpret it Data or Negative Space

Rule-Based NLP Louise's early sessions are pure rule-based NLP: holding up objects, writing words on a whiteboard, establishing one-to-one mappings. These systems are transparent but painfully slow and break the moment language gets ambiguous or figurative. Her whiteboard approach works for concrete nouns like "human" and "Louise," but it cannot handle polysemous words like "offer weapon," and it certainly cannot capture idioms or sarcasm, where the intended meaning has no literal relationship to the words being used. Just as the Sanskrit "desire for more cows" cannot be decoded from its individual words alone, a rule-based system has no mechanism for understanding that "break a leg" means "good luck."

Statistical NLP As more logograms are collected, patterns emerge. Louise recognizes recurring

structures and makes probabilistic guesses based on frequency and co-occurrence. This mirrors statistical NLP: algorithms that find patterns through probability rather than hand-coded rules. They handle variability better but struggle with deep semantic understanding, a limitation reflected in the film's misinterpretations.

2.2 Zero-Resource / Low-Resource Learning

The Crisis: Louise faces a language with no dictionary and no parallel corpus. She builds a lexicon from scratch using **ostensive definition** (pointing and labeling). Imagine learning Japanese with no textbooks, no apps, no bilingual speakers.

NLP Parallel: This mirrors **Zero-Shot Learning** and **Unsupervised Decipherment**. Models like CLIP tackle this through contrastive learning: pairing images with text descriptions to build shared understanding. CLIP does what Louise does: it sees an image and a label together and learns they represent the same concept, building meaning from paired examples rather than pre-existing knowledge.

2.3 Non-Linear Syntax & Parallel Processing

The Crisis: Heptapod B is circular and atemporal. There is no "first word" or "last word." The writer must know the complete meaning before making the first stroke.

NLP Parallel: Early neural networks called **RNNs** processed text one word at a time, carrying a running summary forward. By the end of long sentences, they forgot the beginning (the **linear bottleneck**). The **Transformer architecture** (Vaswani et al., 2017) changed everything with **self-attention**: every word relates to every other word simultaneously, regardless of position. Transformers process language the way Heptapods write it: everything connects to everything, all at once.

3. Communication Methodologies – Research by Chloe

3.1 Neural Transformation: Domain Adaptation

The film's most radical idea is that learning Heptapod B physically rewires Louise's perception. She begins experiencing time non-linearly. In deep learning, this is **fine-tuning**: when a pre-trained model is adapted to a specialized dataset, its internal representations get reorganized to reflect the new domain's worldview. A model fine-tuned on medical journals shifts "discharge" from "firing a weapon" to "releasing a patient." Louise's cognitive transformation is fine-tuning taken to its extreme: the training data is so different from her original model that it restructures the architecture of her thought.

4. Discussion & Theoretical Implications – Research by Chloe

4.1 The Sapir-Whorf Hypothesis & The Alignment Problem

The film portrays the strong form of the Sapir-Whorf hypothesis: language doesn't just influence thought, it **determines** it. From an AI safety perspective, this maps to the **Alignment Problem**: ensuring that an AI system's reasoning aligns with human intentions. Shang's team trained with competitive game data and got competitive interpretations. Louise trained collaboratively and got collaborative answers. Same input, different output, because the training data shaped the model's

reasoning. Your data doesn't just teach a model facts; it teaches it how to see the world.

4.2 The Language-Perception Loop

We identify a **Language-Perception Loop**: in the film, language alters time perception; in AI, training data alters the model's "perception" of truth. Data curation isn't just about accuracy; it shapes the cognitive architecture of the AI. The film argues that genuine understanding requires empathy and shared stakes, not just statistical patterns. Pattern recognition is not comprehension.

5. Conclusion

Arrival offers a rigorous simulation of first-contact linguistics that validates modern NLP strategies. The progression from rule-based bootstrapping to context-aware deep understanding demonstrates why simple translation is insufficient for high-stakes communication. The "weapon" error is a cautionary tale: without robust **Multimodal Grounding** and **Adversarial Alignment**, syntactic accuracy does not guarantee semantic understanding. Getting the grammar right is not the same as getting the meaning right.

The film's central theme, that language shapes thought, maps directly to concerns about AI alignment and the hidden worldviews embedded in training data. The NLP field is still grappling with the question *Arrival* poses: is it enough for a system to produce the right output, or does it need to genuinely understand the input?

6. References

- Boroditsky, L. (2001). *Does Language Shape Thought?* Cognitive Psychology.
- Chiang, T. (2002). *Story of Your Life*. Tor.
- Coon, J. (2020). *The Linguistics of Arrival*. Oxford University Press.
- Devlin, J., et al. (2019). *BERT: Pre-training of Deep Bidirectional Transformers*.
- Pinker, S. (1994). *The Language Instinct*. William Morrow.
- Radford, A., et al. (2021). *Learning Transferable Visual Models (CLIP)*.
- Vaswani, A., et al. (2017). *Attention Is All You Need*.
- Villeneuve, D. (2016). *Arrival*.

7. Appendix: Project Heptapod (Google Colab Notebook) - By DeMarcus

The accompanying Google Colab notebook, **"Project Heptapod: Xenolinguistic Decryption Laboratory,"** provides working Python implementations for every NLP concept analyzed in this report. The notebook contains six interactive sessions:

1. **Session 01: The Weapon/Tool Crisis** | WordNet sense analysis, context-based disambiguation, and BART zero-shot classification demonstrating how context and training bias shape interpretation.
2. **Session 02: Zero-Resource Acquisition** | Xenolinguistic Decryption Unit simulating Louise's vocabulary bootstrapping, plus CLIP multimodal grounding showing how visual-text pairing creates shared meaning.
3. **Session 03: Non-Linear Temporal Syntax** | Linear vs. circular language visualization and procedural Heptapod B logogram generator illustrating simultaneous syntax structure.

4. **Session 04: Transformer Self-Attention** | BERT attention heatmaps and circular chord diagrams demonstrating Heptapod-style parallel token processing.
5. **Session 05: The Sapir-Whorf Effect** | BERT embedding analysis comparing how temporal concepts (past, present, future, simultaneous) cluster in semantic space.
6. **Session 06: NLP Paradigm Evolution** | Radar chart comparing rule-based, statistical, and deep NLP across interpretability, accuracy, and context-handling dimensions.

*See attached: **ARRIVAL_HEPTAPOD_LAB.pdf** for the full notebook output with code, visualizations, and results.*